Hot Plate Assembly

Assembly Instructions:

1. Precautions

- Bond the aluminium plate, the steel base plate, the draft shield and the electronics box to **mains earth**. Check continuity before every first run.
- Put a 10 A MCB and the high-temperature limiter on the live line.
- Do not power a cartridge heater outside its hole. It overheats in seconds and burns out.
- Heat up slowly the first time (see section 6). New heaters and insulation contain moisture and binder that can smoke.
- Do not use acrylic or plastic anywhere near the plate (the draft shield and lid must be metal).
- Aluminium 6061 becomes **soft** after heating to 400 °C. This is fine, because the plate carries no load. Treat the top surface gently, use soft tools, and never clamp on it.
- Do not bend PT100 sheaths sharply or kink heater leads. Do not pull on cables near the heater end.
- Keep the plate's top surface clean. Handle it by the edges, with clean gloves. No oil, hand cream or fingerprints on the top.

2. Tools and consumables

- Multimeter, 500 V insulation tester (if available), IR thermometer
- Dial indicator (0.001 mm) and a granite surface plate
- Drill, reamer set (Ø10.05 and Ø6.1 mm), taps (M4, M6), lathe and mill access, fly-cutter
- Wire cutters/strippers, ferrule crimper, soldering iron, small screwdrivers
- Isopropyl alcohol, lint-free cloth, high-temperature thermal paste, high-temperature ceramic cement
- Safety glasses, heat-resistant gloves, dust mask, fire extinguisher

3. Hole positions in the plate

All positions are measured from the plate centre, with angles counted anticlockwise as seen from the top, starting from a mark ("angle 0") scribed on the plate edge. The heater, sensor and standoff holes are drilled from the **bottom** face, so mark angle 0 on the edge before turning the plate over. Position tolerance: ±0.1 mm.

Feature	Qty	Size	Depth	Position (radius, angle)
Centre-zone heaters	3	Ø10.05 mm (+0.02/0), reamed	40.5 mm, from bottom	r = 25 mm at 90°, 210°, 330°
Edge-ring heaters	6	Ø10.05 mm (+0.02/0), reamed	40.5 mm, from bottom	r = 62 mm at 0°, 60°, 120°, 180°, 240°, 300°
PT100: centre (control)	1	Ø6.1 mm	45 mm, flat bottom	r = 10 mm at 180°
PT100: edge (control)	1	Ø6.1 mm	45 mm, flat bottom	r = 54 mm at 150°
PT100: monitor only	1	Ø6.1 mm	45 mm, flat bottom	r = 38 mm at 20°
Standoff threads	3	M6	10 mm, from bottom	r = 58 mm at 90°, 210°, 330°
Earth bolt thread	1	M4	8 mm, from bottom	r = 66 mm at 270°
Limiter clamp thread	1	M4, drilled sideways	8 mm, at 28 mm above the bottom	on the edge at 30°
Wafer pin holes	3	Ø2 mm (H7)	5 mm, from the top	r = 45 mm at 0°, 120°, 240°

With these numbers, the heater tops end up about 15 mm below the top surface and the PT100 tips end up 10 mm below it. Keep at least 4 mm of metal between any two holes.

4. Assembly steps

A. Prepare the plate

1. **Cut and rough-turn.** Saw the Ø160 mm bar to about 60 mm long. Turn it to Ø151 mm and face both ends to a thickness of 56 mm. The extra material is for the final cut.
2. **Drill and ream** all holes in the table above. Use a flat-bottom drill for the sensor holes so each probe tip can touch the bottom. Deburr every hole.
3. **Stress-relieve the plate.** Do this before the final cut so any warping happens now, not later. Heat the bare plate (no heaters, no sensors) in a workshop furnace to 400 °C at about 100 °C per hour . Hold for 1 hour, then let it cool slowly inside the furnace. Repeat once.
4. **Finish the top and bottom.** Fly-cut the top to a final thickness of 55.0 mm. Face the bottom so it is parallel to the top within 0.05 mm.
5. **Check flatness.** Put the plate on the granite surface plate. Use the dial indicator over a grid of at least 13 points across the central 100 mm. The reading must stay within ±10 µm.
6. **Clean** the plate with isopropyl alcohol. Then break the sharp edges (about 0.5 mm), except do not touch the flat top face.

B. Set the wafer pins

7. **Press the pins in.** Press three Ø2 mm stainless pins into the top holes until each sticks out about 0.4 mm.
8. **Match their heights.** Lap or grind the three pins together until each sticks out 0.15 mm (acceptable range 0.10 to 0.20 mm) . Measure each pin with the dial indicator. The three heights must agree within about 5 µm. A bigger gap at one pin means less heat reaches the wafer there.

C. Fit the heaters and sensors

9. **Dry-fit.** Slide each heater and probe into its hole by hand to check the fit, then remove it. If one will not go in freely, do not force it.
10. **Install the heaters.** Make sure each hole is clean and dry. Insert all 9 heaters fully, with the leads pointing out of the bottom. If required use thermal paste or otherwise, insert them clean and dry.
11. **Install the PT100 probes.** Lay a thin film of high-temperature thermal paste on each probe and slide it in until the tip touches the bottom of its hole and then it will slide out automatically. Hold each one in place with a small blob of high-temperature ceramic cement at the hole mouth, keeping the cable clear of it. Let the cement cure as its label says.
12. **Label every wire** at both ends ("C1 to C3" for centre heaters, "E1 to E6" for edge heaters, "PT-C", "PT-E", "PT-M" for the sensors).

D. Base, standoffs and insulation

13. **Make the base plate.** Cut a 4 mm steel plate about 250 × 250 mm. Cut 3 slots, each 6.5 mm wide and 12 mm long, at the 3 standoff positions. Point each slot along the line from the plate centre outwards, so the plate can expand freely.
14. **Cut the insulation.** From the 25 mm ceramic board cut one disc, Ø180 mm (bottom layer), and two rings, outer Ø180 mm and inner Ø152 mm (two layers make a 50 mm wall). Wear the dust mask. Seal the cut faces with a ceramic-fibre rigidizer, or wrap the outside in aluminium foil tape, to keep fibres from coming loose.
15. **Drill the lead holes.** Place the bottom disc on the base plate and set the aluminium plate on top. Mark where each heater lead and sensor stem comes out (12 places). Drill Ø12 mm holes through the disc and the base plate at those marks.
16. **Mount the plate on standoffs.**
    1. Lay the disc on the base plate with the holes lined up.
    2. Stand three thin-wall stainless tubes (Ø10 × 1 mm, 28 mm long) over the slots.
    3. Lower the plate onto the tubes so the M6 threads line up. Feed the leads and sensor stems through their holes.
    4. Insert an M6 bolt and flat washer through each slot from below and tighten just until the tubes stop wobbling, then confirm the base can still slide a little in its slots.
    5. Check the plate is level to better than 0.1 mm across 100 mm. A tilted plate changes the wafer gap. Add thin stainless shims under a foot if needed.
17. **Fit the side ring.** Place the two rings around the plate on top of the bottom disc. Leave about 1 mm clearance to the plate and keep the ring top at or just below the plate's top surface, never above it. Hold the rings with a stainless band clamp.

E. Limiter and earth

18. **Fit the limiter.** Clamp the high-temperature limiter to the plate's side at 30° with a small stainless bracket and an M4 screw. Add a thin film of high-temperature paste where it touches the metal. Route its wires out through a notch in the bottom of the ring. Use only wire and sleeving rated for the limiter's temperature near the plate.
19. **Fit the earth bond.** Fix a ring lug to the M4 hole in the bottom face with a star washer. Run 1.5 mm^2 green/yellow wire to the mains earth terminal. Also bond the base plate and draft shield to earth. With the multimeter, check that the plate to earth pin reads less than 1 Ω.

F. Electronics and wiring

20. **Mount the electronics** in the project box at least **30 cm** away from the plate, out of its heat path. Mount each SSR on its heat sink with thermal paste. Keep mains terminals separated from the low-voltage side by a barrier or at least 6 mm of air.
21. **Wire the low-voltage side.** Use the pin table below.

Connection	**Pico pin**
MAX31865 SCK (all 3)	GP18
MAX31865 MOSI/SDI (all 3)	GP19
MAX31865 MISO/SDO (all 3)	GP16
MAX31865 chip select: centre	GP17
MAX31865 chip select: edge	GP20
MAX31865 chip select: monitor	GP21
MAX31865 power	3V3 and GND
Centre SSR driver	GP14
Edge SSR driver	GP15

22. **For each SSR driver**: Pico pin → 1 kΩ → BC547 base. Add a 10 kΩ resistor from the base to GND, so the SSR is **off** when the Pico is not driving it. BC547 emitter → GND, collector → SSR input "−", SSR input "+" → 5 V.

    PT100 wiring (4-wire): with the multimeter, find the wire pairs. Wires of the same colour read close to 0 Ω, and red-to-white reads about 110 Ω at room temperature (28 °C). Connect each pair to the matching F+/RTD+ and F−/RTD− terminals on its MAX31865, and set the board's solder jumpers for 4-wire. Check the board has a 430 Ω reference resistor. Use shielded cable and connect the shield to GND at the MAX31865 end only.

23. **Wire the mains side.** Use 1.5 mm^2 cable and ferrules.
    - Live: plug → 10 A MCB → high-temperature limiter → input terminal of **both** SSRs.
    - Each SSR output goes to its heater group's live: centre SSR → 3 centre heaters in parallel; edge SSR → 6 edge heaters in parallel.

- Neutral goes straight to all heaters through a terminal block.
- Earth goes to the plate, base plate, draft shield and box.
- Join the heater leads to the mains wire in the cool zone, at least 15 cm below the base plate, using terminal blocks and high-temperature (fibreglass) wire for any part that gets warm.

G. Shield and lid

23. **Fit the lid.** Cut a lid from 1 mm aluminium or stainless sheet, a little wider than the wafer. Hang it from the draft shield on stainless rods so it sits about 15 mm above the wafer .
24. **Fit the draft shield** around the plate with at least 5 cm of clearance on all sides. Make sure it cannot touch the plate.

5. Checks before power

Check	**Expected result**
Each heater's resistance	about 330 Ω (230 V^2 ÷ 160 W), within roughly 315 to 365 Ω
Centre group (3 in parallel)	about 110 Ω
Edge group (6 in parallel)	about 55 Ω
Insulation resistance, heater to plate (500 V tester)	more than 1 MΩ. New heaters can read low until the moisture is driven out. If low, do the low-power dry-out first.
Plate to earth pin	less than 1 Ω
Three PT100 readings at room temperature	within about 1 °C of each other and of the room
SSR inputs with the Pico unplugged	off (no heater power)

6. First power-up precautions

Ventilate the room because the first heating can smoke.

1. **Dry-out at 100 °C.** Hold the plate at 100 °C for 30 to 60 minutes This drives moisture out of the heaters and binder out of the insulation.
2. **Step up** to 150, 250, 350 and then 400 °C. Hold about 30 minutes at each step. At each step check: both zones climb together, the limiter and cable temperatures stay moderate (IR thermometer on the base plate and wires) and the shield stays cool.
3. **Tune the PID.** Run autotune on each zone at about 200 °C, then fine-tune at 400 °C.
4. **Check the limiter and software limit.** With the heaters unpowered, confirm the program turns both outputs off when you simulate a sensor fault (for example, unplug one probe).
5. **Cool down slowly,** then re-check the top surface flatness with the dial indicator. Repeat the heat cycle once and measure again. Record both results.

7. Quick fault guide

Problem	**Likely cause**	**What to do**
MAX31865 reports a fault	Wrong wire pair, loose connection, or wrong reference resistor	Re-check the wire pairs and the board's 4-wire jumpers
One zone heats much slower	A heater is open, or a wire is loose	Measure each heater's resistance
Edge is colder than centre	Edge zone power too low	Raise the edge/centre power ratio, then re-map the wafer
Temperature swings around the setpoint	PID not tuned	Re-run autotune, lengthen the SSR time window slightly
Smoke or burning smell	First-heat binder or moisture, or a wiring problem	Power off. If it keeps happening, find the source before restarting
Limiter trips below 400 °C	Limiter tolerance or the wrong variant	Check its rated switching temperature and the sensor readings